How Often a Frozen Mix Recurs across Unused Initializations:
A Closed Three-Seed Count, Not a Population Estimate

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Abstract

Does a prospectively frozen token-share mixture, already measured once on a historical initialization, recur on unused parent initializations that were named before any P5 parent, child, validation BPB, or test BPB existed? This question was preregistered as AsPredicted #307836 after P4 unblinding. P5 is a post-P4, prospectively preregistered closed panel of the frozen P4 apparatus. It does not amend AsPredicted #306780, #306935, #307342, or #307591. C3 is not P3 B3. P4 seed 0 is a previously released historical cell, not a P5 confirmatory cell. Three unused parent-init seeds $\{1, 2, 3\}$ were executed in that order with no replacement and no early stop. Each seed trained a fresh Tagalog parent on WikiText-TL-39 with Karpathy's nanochat trainer pinned at commit `92d63d4`, the carry-forward P4 tokenizer, $T = 2048$, and NVIDIA CUDA A40. After P0-T PASS, d20 was frozen as immutable $C0_s$. Matched d20 continuations compared C1 extra Tagalog, C2 pure English, and C3 the P4-frozen mix at $q_{\mathrm{TL}} = 0.50$ for $D_{\mathrm{phase2}} = 19{,}267{,}584$ tokens ($N = 294$). Per-seed co-primary estimands are $R^{(s)}_{\mathrm{TL}} = \mathrm{TL}(C3_s) - \mathrm{TL}(C2_s) \leq -0.01$ and $A^{(s)}_{\mathrm{EN}} = \mathrm{EN}(C3_s) - \mathrm{EN}(C1_s) \leq -0.01$. The primary panel result is the count table: eligible $K_{\mathrm{elig}} = 3$; both criteria met $k_{\mathrm{both}} = 3$; only-$R = 0$; only-$A = 0$; neither $= 0$; ineligible parent $= 0$. In all three predeclared unused initializations, the P4 token-share mixture pattern recurred. This is a recurrence count in a closed panel, not a confidence interval, not a population effect, and not an independent confirmation of P4 as a law.

Keywords: unused initializations; recurrence count; closed panel; token share; Tagalog; English; bits-per-byte; nanochat; preregistration; post-P4.

arXiv categories: cs.CL, cs.LG.

Registration and deposits (P5): AsPredicted #307836 (https://aspredicted.org/k6ib64.pdf); ResearchBox pending (new box, not #8869); study record https://github.com/pageman/nanochat-filipino/tree/main/docs/p5; Hub https://huggingface.co/pageman/nanochat-filipino-p5-p4-multi-seed (weights deferred: C0+C1+C2+C3 per eligible seed together). Run ID: `p5-20260823T160632Z-439d1de5`.

1 Introduction

Pajo [1] measured equal-exposure Tagalog BPB across nanochat depths on WikiText-TL-39 (AsPredicted #306780). Pajo [2] asked the forward continual-pretraining question from an English parent (AsPredicted #306935). Pajo [3] asked the reverse from a fresh Tagalog parent, with a 50/50-*document* B3 arm (AsPredicted #307342). Pajo [4] locked source-content token share at $q_{\mathrm{TL}} = 0.50$ after a fresh Tagalog parent (AsPredicted #307591) and observed both co-primary inequalities in *one* initialization (P4 seed 0).

P5 was designed after those released P4 findings were known. It is therefore a prospectively preregistered **post-P4** multi-seed panel of the frozen P4 apparatus [5]. It is not an amendment, correction, rescue, replication, or independent confirmation of P4. It is not "P4 confirmed as a law." C3 remains the P4-frozen token-share mix; C3 is not P3 B3. P4 seed 0 is historical disclosure, not a P5 confirmatory cell.

The confirmatory question, locked before any P5 confirmatory BPB existed, is:

> For each unused parent-init seed $s \in \{1, 2, 3\}$ executed in that order: train a fresh Tagalog parent, apply P0-T, freeze d20 as $C0_s$ if PASS, and continue it on C1 extra Tagalog, C2 pure English, and C3 the P4-frozen mix. Classify each eligible seed as `both`, `only-R`, `only-A`, or `neither` at $\delta = 0.01$. Report the closed-panel counts. Do not early-stop. Do not replace. Do not unblind per seed.

This paper contributes: (i) three unused Tagalog parents with P0-T PASS, each frozen as its own $C0_s$; (ii) eighteen child validation cells plus three descriptive C0 English cells, sealed before one C3-only secondary test per eligible seed; (iii) one panel Gate X whose primary result is the count table $k_{\mathrm{both}} = 3$ of $K_{\mathrm{elig}} = 3$.

2 Related work

Cruz and Cheng constructed WikiText-TL-39 [6]. Sequential interference is classical [12, 13]. LLM continual-pretraining work often continues an English or multilingual base [10, 11, 9]. P5 stays inside WikiText-family text, the nanochat depth dial, and BPB rather than CORE or chat [7]. Equal token budgets follow the family so stream identity is not confounded with compute [14, 15, 16]. WikiText established clean Wikipedia LM evaluation in English [8]. BPB remains comparable across vocabularies [17, 18]. Official P5 BPB is mean token NLL divided by $\ln 2$ times mean UTF-8 bytes per token, full split, $T = 2048$, BOS-best-fit. In-loop trainer BPB is not confirmatory. Preregistration reduces undisclosed flexibility [19, 20]. A closed count of unused initializations is not a random-effects model and is not a population rate.

3 Methods

3.1 Registration and authority

AsPredicted #307836 governs [5].

- PDF: `https://aspredicted.org/k6ib64.pdf`
- PDF SHA-256:
  `439d1de5ff9fd18e466f33192c5ac9c5c36b020ca72942ae218b9e69a8f5bbf3`
- Gate-plan SHA-256 at filing:
  `d51115aade9c0b1fb8698eaa33540db2d75b2b27765aaaad1bf14b13b0132092`
- Pre-filing addendum SHA-256:
  `839fcaa3dd6e94bd9546df4880a5892851a54ea31743cb0359adc8faebbe9258`

A new ResearchBox (not #8869) is the intended deposit; it is pending and is not made public by this paper. Authority: filed PDF $\gg$ hash-pinned plan/addendum $\gg$ `LOCK.json` $\gg$ dated deviations $\gg$ this manuscript. The study does not amend #306780, #306935, #307342, or #307591. C3 is not P3 B3. P5 is computationally independent of P4 in parents, children, ledgers, and P5 outcomes, and is not outcome-independent of P4, because released P4 seed-0 findings informed the design.

3.2 Sources, tokenizer, and hosts

Tagalog train/val/test documents are the frozen WikiText-TL-39 P1.1-eligible splits. English train/val/test documents are the frozen WikiText-103 P2-eligible splits. The tokenizer is the carry-forward P4 pair: `tokenizer.pkl` `04436b854e0841025a3dd2b46baaeeea07a7ccc252e9f99a19171306f00bc5a8` and `token_bytes.pt` `a5dbc1c88f6292696108263072d77115718cc2d8357f7ad4859adfa517cc2132`. P1.1/P2/P3/P4 weights were not loaded as parents. Official GPU gates used NVIDIA CUDA A40. Official evaluation used `scripts/p5/evaluate_bpb.py`.

3.3 Panel seeds and parent initialization

The closed seed list is $\{1, 2, 3\}$ in that order. Seed 0 was used only for one nonconfirmatory d4 smoke and is not a panel cell. Wrapper Python/NumPy seed is 424242. Immediately before `GPT.init_weights()` for each parent d8 and d20, the wrapper set `torch.manual_seed(`s`)` and `torch.cuda.manual_seed(`s`)`. Distinct initial-state SHA-256 values were recorded before parent training.

3.4 P0-T and C0 freeze

P0-T compared full-split Tagalog validation BPB of d8 and d20 TL0 against a seed-matched untrained initialization and a byte-unigram floor at $\delta_{\mathrm{P0T}} = 0.01$. Official P0-T status is CUDA-only; safe public output before Gate X was restricted to PASS, BLOCKED, or TECHNICAL_STOP. After PASS, d20 was copied to an immutable $C0_s$ directory with `additional_train_tokens`=0. Smoke and d8 checkpoints are not C0. A BLOCKED seed would have been reported as ineligible, not replaced; that branch was not triggered.

3.5 Children and C3 identity

C1, C2, and C3 each continued the same frozen $C0_s$ for $N = 294$ steps, $B = 65{,}536$, $D_{\text{phase2}} = 19{,}267{,}584$ tokens, serial R$\to$S$\to$T, `load_optimizer=False`, child peak learning rate $0.3\times$ the parent peak, warmup 14, terminal checkpoint only. C1 used extra Tagalog. C2 used WikiText-103 raw English train. C3 used the P4-frozen mix (not reshuffled): source-content quotas 9,633,792 / 9,633,792, interleave seed 42, mix-manifest SHA-256 `f203c615266bc8c33c358c1de399715791cae33536a9743c8a6bf8cd543cb107`. Packed C3 shard hashes and `language_origin_mask` SHA-256 `140e174a427a7ddf2126553c53352ec049f72fbed475e2404cd4ef122b309c46` were verified; matching source JSONLs or the manifest alone is not sufficient C3 identity.

3.6 Lockbox, analysis, and tests

Validation BPB was lockboxed until one panel Gate X. Each eligible seed's Gate U sealed six child cells plus descriptive C0 English with `test_access[`s`]`=0 and no contrasts computed at seal. Policy authorized one C3-only secondary test event after U (Gate V). C1 and C2 were not tested. Contrasts use sealed validation cells; tests are secondary and excluded from classification. Equality at $-\delta$ counts. No composite score, mean, confidence interval, p-value, or random-effects model.

4 Results

All three seeds completed the filed path through V. P0-T status is **PASS** for seeds 1, 2, and 3. Table 1 reports Tagalog validation BPB of the trained parents against the two floors. Table 2 reports the sealed child cells. Table 3 reports per-seed contrasts and class. Table 4 is the primary panel result.

Table 1: P0-T Tagalog full-split validation BPB (six decimals). Unigram floor is 4.453225 for all seeds.

Seed	d8 trained	d8 untrained	d20 trained	d20 untrained
1	1.026063	3.289120	0.713846	3.288977
2	1.025926	3.289165	0.866993	3.289237
3	1.025522	3.289167	0.835588	3.289509

Table 2: Full-split validation BPB after terminal d20 checkpoints. Six-decimal reporting as filed. C0 English is descriptive and excluded from the contrasts.

Seed	Arm	Tagalog val BPB	English val BPB
1	C0 (frozen parent)	—	2.602629
1	C1 extra Tagalog	0.307864	2.972254
1	C2 pure English	1.947782	1.248221
1	C3 token-share mix	0.733086	1.417082
2	C0 (frozen parent)	—	2.566109
2	C1 extra Tagalog	0.503637	2.874332
2	C2 pure English	2.222498	1.319657
2	C3 token-share mix	0.870142	1.486114
3	C0 (frozen parent)	—	2.549180
3	C1 extra Tagalog	0.449775	2.854149
3	C2 pure English	2.171966	1.296541
3	C3 token-share mix	0.837820	1.467922

Table 3: Per-seed co-primary contrasts from sealed validation cells. Filed criterion ≤ -0.01; equality counts.

Seed	R_{TL}	A_{EN}	Class
1	−1.214696	−1.555172	both
2	−1.352356	−1.388218	both
3	−1.334145	−1.386228	both

Under this preregistered token-share-locked mixture, P5 seed 1 observed lower Tagalog BPB than pure English continuation and lower English BPB than extra Tagalog continuation in the

Table 4: Primary panel result: counts in the closed $K = 3$ panel. This table *is* the panel claim.

Cell	Count
Eligible seeds K_{elig}	3
Both criteria met k_{both}	3
Only R_{TL}	0
Only A_{EN}	0
Neither	0
Ineligible parent	0

stated apparatus. This is one initialization in a predeclared panel; not a CI; C3 is not P3 B3; P5 does not amend #307591.

Under this preregistered token-share-locked mixture, P5 seed 2 observed lower Tagalog BPB than pure English continuation and lower English BPB than extra Tagalog continuation in the stated apparatus. This is one initialization in a predeclared panel; not a CI; C3 is not P3 B3; P5 does not amend #307591.

Under this preregistered token-share-locked mixture, P5 seed 3 observed lower Tagalog BPB than pure English continuation and lower English BPB than extra Tagalog continuation in the stated apparatus. This is one initialization in a predeclared panel; not a CI; C3 is not P3 B3; P5 does not amend #307591.

Across the $K = 3$ predeclared unused initializations, both co-primary P4 criteria were met in $k_{\text{both}} = 3$ of $K_{\text{elig}} = 3$ eligible seeds ($k_{\text{inelig}} = 0$ ineligible). In all K predeclared unused initializations, the P4 token-share mixture pattern recurred. This is a recurrence count in a closed panel, not a population effect, not a confidence interval, and not an optimal-ratio result. P4 seed 0 is a previously released historical cell, not a P5 confirmatory cell; C3 is not B3; tests are secondary; no silent amendment of #307591.

A measured reduction in the P3-style relative Tagalog retention cost within this frozen P5 replica of the P4 apparatus, not a general mitigation of catastrophic forgetting. That optional sentence is per eligible seed (all three met both criteria) and does not replace Table 4 as the primary panel claim. It does not appear in the title.

One authorized C3-only secondary test event per eligible seed reported English / Tagalog test BPB 1.426163 / 1.201608 (seed 1), 1.498279 / 1.148789 (seed 2), and 1.479601 / 1.146587 (seed 3). Tests do not alter sealed contrasts and are not co-primary. P4 Gate V numbers are not reused.

5 Discussion

C3 sits between C1 and C2 on both languages in every eligible seed, as a token-share trade-off rather than a claim that $q_{\text{TL}} = 0.50$ is optimal. Recurrence in all three unused initializations is a count, not a law. P5 does not independently confirm P4. P5 does not confirm P3. P5 does not show that P3 B3 was “fixed.” Mechanism (schedule topology, revisit, optimizer state) is reserved for later filings. Byte share of the packed mix remains descriptive; 50/50 source-content tokens is not 50/50 documents and is not 50/50 UTF-8 bytes.

6 Limitations

$K = 3$ unused initializations; one tokenizer; WikiText-family text only; no chat, SFT, CORE, FilBench, or larger-model transfer. Confirmatory GPU work used a rented NVIDIA A40. Raw test JSONL remains restricted. Hub weight release, if any, must ship C0+C1+C2+C3 together per eligible seed (or all eligible seeds together) with tokenizer and meta; C3 alone is prohibited; a single seed's C3 is not "the P5 model"; P4 Hub IDs must not be overwritten. Byte-balanced mix remains a separate filing (P6-B). A still-larger panel would be a new filing, not "P5 plus seed 4."

7 Deviations

No protocol stop. No break-glass. No additional confirmatory validation or test eval after Gate X. Seed 2 Gate S wrote a terminal C2 checkpoint, after which the host died (volume quota then GPU lockout). The official terminal checkpoint was accepted after reload and metadata step verification (`step_source=meta`); C2 was not retrained into a filled directory. The subsequent pod restart did not rerun Gate H or any completed parent. Frozen English train JSONL contains intra-split duplicate rows (28,472 rows / 28,232 unique); they were not dropped. Cross-split overlap was 0.

Availability

- AsPredicted #307836: https://aspredicted.org/k6ib64.pdf
- ResearchBox: pending new box (not #8869); FOR PEER REVIEW when deposited; not Make Public
- Study record (GitHub):
  https://github.com/pageman/nanochat-filipino/tree/main/docs/p5
- Code and audit trees (GitHub):
  https://github.com/pageman/nanochat-filipino
  P5-only: `scripts/p5/`, `docs/papers/p5-multi-seed-p4/`, `docs/run-cards/p5/`, `results/p5/`, `docs/hub/p5-p4-multi-seed/`, `manifests/p5/`
- Weights (Hugging Face Hub):
  https://huggingface.co/pageman/nanochat-filipino-p5-p4-multi-seed
  (deferred; per eligible seed C0+C1+C2+C3 together; never C3 alone; never write onto the P4 Hub ID).

Held-out test text is not redistributed. Host credentials are not published. GitHub holds protocol, receipts, sealed JSON, and paper; Hugging Face holds optional weight bundles.

Ethics

Public Wikipedia-derived corpora. No human-subjects experiment. Any ResearchBox remains passcode-protected for peer review and is not made public by this paper.

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A Selected hashes

- Pin / nanochat: `92d63d4e8bb4df75c3b71618f31ddde2378b2bcd`
- AsPredicted PDF: `439d1de5ff9fd18e466f33192c5ac9c5c36b020ca72942ae218b9e69a8f5bbf3`
- Gate plan: `d51115aade9c0b1fb8698eaa33540db2d75b2b27765aaaad1bf14b13b0132092`
- Addendum: `839fcaa3dd6e94bd9546df4880a5892851a54ea31743cb0359adc8faebbe9258`
- Mix manifest: `f203c615266bc8c33c358c1de397715791cae33536a9743c8a6bf8cd543cb107`
- Evaluator: `78e77d9761fa4473344abd4773129e56f04561e615bdbcfad144ab8730ded977`
- Seed 1 C0 / C1 / C2 / C3: `c5bf0f495f7cb296374105c62b5946aa30cfb33fa86c58c08434c37862631ac6` / `8d474aa88a250895dd410b0027277dd19a2b8f41a77544902ec4bbc774c33f14` / `f9701fae3277f5c379d69d97dd3ba9b706324bf72660731b0a4fbb34371e628e` / `f7c080048d4311863836fb6cf2c06d9d555514bcce7e30a5d8b819aa0dd15252`
- Seed 2 C0 / C1 / C2 / C3: `e4c4cc7bcdf5033a97f9eedd97f7818d3c8e3ddca2b46fffc8b8c78d6137c4b1` / `aaee36fd5ceb53c2e92138eb2dfa8b169bd000caf5b804d9c8645d6ad56c4f22` / `e75b4c61bf3dd2cd661d931dfc63526c810dd96bf8db2fcac0f2cadb5e29c102` / `0801f1437b334a68ee155623111ecba2dc03cfc49c46e8630f023c41aa21ac77`
- Seed 3 C0 / C1 / C2 / C3: `7643b12645f487b4f38034f14985c624804ac43534cfafc723bae83edc4c3c26` / `edfabbf8dbb38aefbfd61850f30efba40d8d77c8dffc02ce4db403969787b66f` / `e3cd58a5fa549b62c60c568a551f3d471df5b77698208a4727c0a52266d3f58a` / `1759110154b2a4176e584bcd675d4b21d10cda0f9936e8f6a188aa32ba8f2ab7`
- U seals 1 / 2 / 3: `4f843ca7b5bb6935baf0ad4e8d636accb49c88d43ce6b32f5fb6f0aaeb170874` / `75c5ab56804c937a1436d538c3b4350830bd239924de76e4a877aa0988edfea0` / `df15915f2c5855691b6e0a4a5b9c0e62aa9229b8a246d932a7f4926074561cf5`

B GitHub versus Hugging Face

GitHub `pageman/nanochat-filipino` is the study record: scripts, paper, lock, ledgers, run-card receipts, sealed JSON, and Hub *documentation* (model cards without weights). Hugging Face `pageman/nanochat-filipino-p5-p4-multi-seed` is the optional weight deposit: per eligible seed, C0+C1+C2+C3 plus tokenizer and evaluation JSON. Never C3 alone. Never a single seed's C3 as "the P5 model." Never write onto P1.1/P2/P3/P4 Hub IDs. Raw test JSONL, passcodes, SSH keys, optimizer states, and HOST operator cards belong in neither public tree.

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